Master Marine Physics, 2nd year
Geophysical Fluid Dynamics
Exercise sheet # 2 : Geostrophic balance, thermal wind balance, potential vorticity, quasi-geostrophic motion

1 Comparison of hydrostatic and dynamical pressures

Compare the horizontal difference of dynamic pressure associated with the Gulf Stream (compute it with the geostrophic balance and the Gulf Stream characteristics, $U = 1m/s, L = 40km, f = 10^{-4}s^{-1}$) to the hydrostatic pressure at the bottom of the Gulf Stream (at depth $H = 500m$). Conclude on the possibility to measure the dynamic pressure with pressure gauges.

2 Mid latitude winds

What is the geostrophic velocity of westerlies between the Azores anticyclone (1030 hPa) and the Iceland depression (980 hPa) distant of about 3000 km ? Use f_0 at 45°N.

3 Thermal wind relation

The Gulf Stream separates the 10°C water of the Blake plateau from the 18°C water of the Sargasso Sea. Its depth is 500 m, its surface velocity is 1 m/s (and the velocity at a depth of 500 m is nearly null). What is its half width L assuming that the coastal and offshore waters are separated by about 2L) ?

4 Cyclogeostrophic balance of a vortex

The shallow-water equations on the f-plane ($f = f_0$) are written in polar coordinates

$$\partial_t v_r + v_r \partial_r v_r + \frac{v_\theta}{r}\partial_\theta v_r - \frac{v_\theta^2}{r} - f_0 v_\theta = \frac{-1}{\rho_0}\partial_r p$$

$$\partial_t v_\theta + v_r \partial_r v_\theta + \frac{v_\theta}{r}\partial_\theta v_\theta + \frac{v_r v_\theta}{r} + f_0 v_r = \frac{-1}{\rho_0 r}\partial_\theta p$$

$$\partial_t h + v_r \partial_r h + \frac{v_\theta}{r}\partial_\theta h + h\,(\frac{1}{r}\partial_r(r v_r) + \frac{1}{r}\partial_\theta v_\theta) = 0$$

1) Show that, under the ypothesis of a circular flow ($v_r = 0, v_\theta(r), h(r), p(r)$), the dynamical equations reduce to the cyclogeostrophic balance, relating diagnostically the horizontal pressure gradient and the orthoradial velocity.
2) Interpret this equilibrium. Plot v_θ with respect to $\partial_r p/\rho_0$.
3) Caracterise vortex structures and the balance of forces in each region of the (pressure gradient, velocity) plane.

5 Current crossing (or not) a topographic step

(From Benoit Cushman-Roisin's book)
We consider a homogeneous ocean, bounded above by a rigid lid and below by a step topography (see figure 1). The ocean is bounded by a northern coast (at $y = 0$). Along this coast, a zonal, laterally sheared current $U(y) = U_1(1+y/L_1)$ propagates eastward ($U_1 < 0$). This current has constant shear U_1/L_1 and finite width L_1. The flow obeys the 2D divergent (3D non divergent), but stationary, shallow-water equations, on the f-plane.

Show that
1) there exists a transport streamfunction given by $hu = -\partial_y\psi, hv = \partial_x\psi$. From the conservation equation of potential vorticity Q, show that $J(\psi, Q) = 0$, where J is the Jacobian operator. Compute the potential vorticity for $y \in [-L_1, 0]$. Conclude that the external boundary of the flow is a transport streamline and thus that no clow can cross it.
2) Compute the current transport
3) The current encounters the topographic step. The depth changes abruptly from H_1 (constant) to H_2 (constant, but smaller). The new structure of the

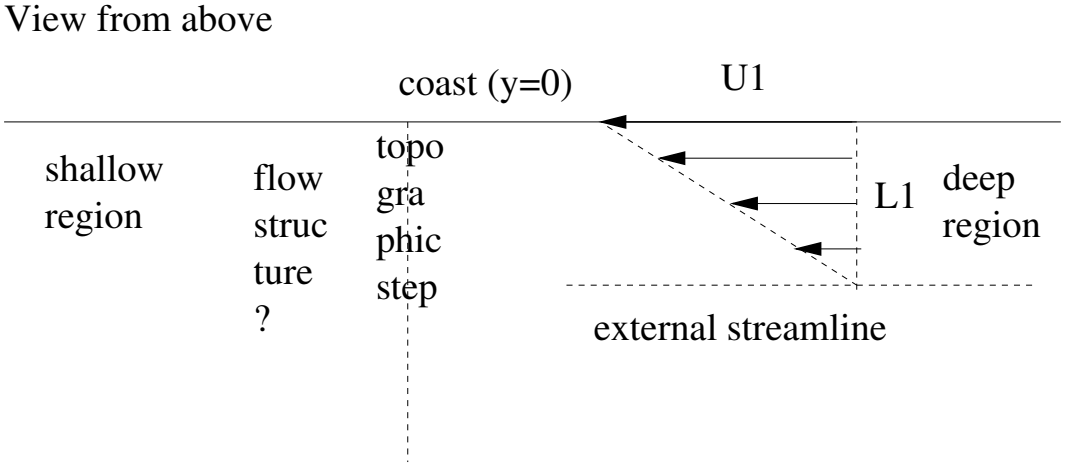


FIGURE 1 – Current climbing on a step

current (if it climbs on the step) will now be computed. Use the Lagrangian conservation of potential vorticity and the conservation of current transport to compute the structure of the current once it has climbed on the step, compute its new width and maximum current.
4) Numerical application : the values are $H_1 = 200m$, $H_2 = 160m$, $f_0 = 10^{-4}s^{-1}$, $U_1 = 0.5m/s$, $L_1 = 10km$. Compute U_2 and L_2.
5) What would happen if $H_2 = 100m$?
6) What is the lateral difference in dynamic pressure across the jet upstream and donwstream of the step (for $H_2 = 160m, \rho = 10^3 kg/m^3$) ?

6 Exit of a channel in a Bay

A channel 200m deep, 300m wide, empties in a bay. Compute the resulting current along the coasts. Use f-plane, 2D divergent, steady shallow-water dynamics (homogeneous ocean with rigid lid and step like, flat bottom). Use $f_0 = 10^{-4}s^{-1}$.

7 Current on a bottom slope

A steady current with uniform potential vorticity $q_0 = 5\,10^{-7}m^{-1}s^{-1}$ and total transport $T_0 = 4\,10^4 m^3/s$ flows on a uniform bottom slope ($s =$

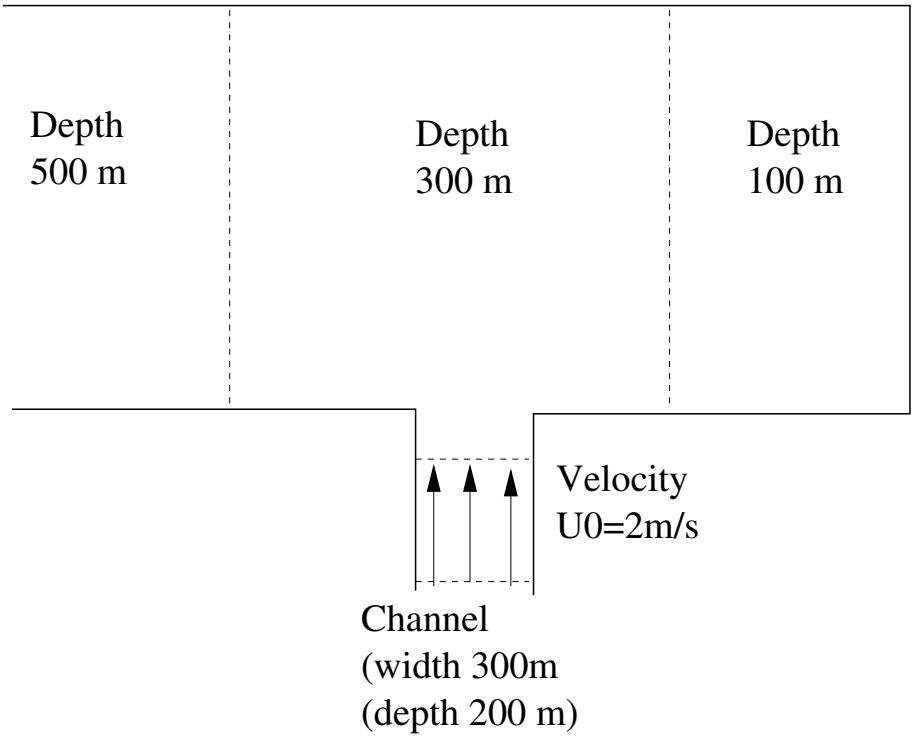


FIGURE 2 – Current entering a bay from a channel

$1m/km$). Show that the current profile is parabolic. What is the width of the current and its depth at maximum velocity ?

8 Current crossing a mid ocean ridge

A uniform zonal steady current, encounters a mid ocean ridge perpendicularly. The local height od the ridge is $h(x)$. Using potential vorticity conservation in the quasi-geostrophic model, show that the current is deflected southward.